

Question Number	Answer	Additional guidance	Mark
3(a)(i)	The only correct answer is C – plant cells and prokaryotic cells <i>A is incorrect because prokaryotic cells also have a cell wall</i> <i>B is incorrect because animal cells do not have a cell wall</i> <i>D is incorrect because animal cells do not have a cell wall</i>		(1)

Question Number	Answer	Additional guidance	Mark
3(a)(ii)	Drawing (2): • {double (nuclear) membrane / nuclear envelope} obvious • nuclear pores shown and (1 or more) nucleoli present Labels (max 2): • (nuclear) envelope / <u>double</u> membrane / {<u>inner</u> / <u>outer</u>} (nuclear) membrane • (nuclear) pore • nucleolus • correct reference to chromatin / nucleoplasm	if one incorrect label, max of 1 label mark if two incorrect labels then no label marks can be awarded Ignore DNA	(4)

Question Number	Answer	Additional guidance	Mark
3(b)(i)	An explanation that includes the following points: - the two {layers (of the membrane) /membranes} are very close together / there is a very small distance between the {two layers / membranes} (1) - therefore, the resolution of a light microscope is not high enough (1)	Accept the bilayer structure is very close together Ignore the bilayer is too {small / thin}	(2)

Question Number	Answer	Additional guidance	Mark
3(b)(ii)	A calculation showing the following steps: - length of XY measured correctly +/- 1mm (1) - calculation of actual size (1) - correct answer given in μm (1)	Mark the answer on answer line first <u>Example of calculation</u> 26 mm / 2.6 cm 26 ÷ 200= 0.13 (mm) = 130(μm) Correct answer scores full marks	(3)

Question Number	Answer	Additional guidance	Mark
3(b)(iii)	A description that makes reference to the following: • form spindle {fibres / poles} in {prophase / mitosis / meiosis / cell division} (1) • to separate {chromosomes/ chromatids} (1)	Ignore DNA / genetic material	(2)